

Probing the Relationship Between Bulk and Local Environments to Understand Impacts on Electrocatalytic Oxygen Reduction Reaction

Brianna N. Ruggiero^a, Kenzie M. Sanroman Gutierrez^a, Jithin D. George^b, Niall M. Mangan^b,
Justin M. Notestein^a, Linsey C. Seitz^{a,*}

^a *Department of Chemical and Biological Engineering*

Northwestern University, Evanston, IL, 60208

Email: briannaruggiero2023@u.northwestern.edu, j-notestein@northwestern.edu,

*linsey.seitz@northwestern.edu**

^b *Department of Engineering Sciences and Applied Mathematics*

Northwestern University, Evanston, IL, 60208

Email: jithingeorge93@gmail.com, niallmm@gmail.com

Abstract

Local catalyst environments fluctuate during electrochemical reactions which significantly influences reaction efficiency and selectivity. However, elucidating fundamental relationships between local environment and electrocatalytic performance is challenging due to the difficulty in probing this local region. Here, we study the impact of electrolyte composition on the kinetics and mechanism of the oxygen reduction reaction (ORR) for a carbon catalyst. Using a rotating ring-disk electrode equipped with an iridium oxide pH probe, we monitor changes in local pH during ORR. We confirm that local pH changes significantly for near-neutral bulk pH and find that different cations provide variable pH-buffering, thereby modulating overall magnitude and onset of local pH changes. Overall, local pH more strongly dictates changes in mechanisms and performance, while the cation identity modulates local pH to influence these trends. This work highlights the importance relating local pH to electrocatalytic performance to further improve understanding of complex electrochemical processes.

Keywords

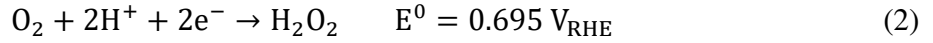
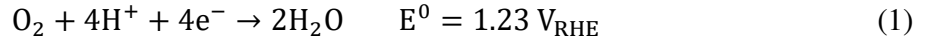
local pH; oxygen reduction reaction (ORR); carbon electrocatalyst; cation effects; hydrogen peroxide; electrocatalysis

1. Introduction

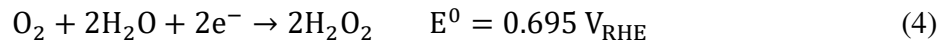
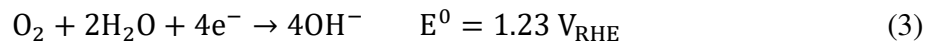
The electrochemical oxygen reduction reaction (ORR) plays an important role in the development of sustainable energy conversion and storage technologies.¹ It can convert O₂ to H₂O via a 4e⁻ pathway for fuel cell applications or to hydrogen peroxide (H₂O₂) via a 2e⁻

pathway, providing a sustainable and efficient method of producing this environmentally friendly oxidant chemical.² H₂O₂ is widely used in a variety of industrial processes including textile bleaching, environmental protection, and chemical production. Recently, the global demand for H₂O₂ has substantially increased, resulting from the global public health safety challenges, and is expected to grow further to 5.7 million metric tons by 2027.^{3, 4} At present, H₂O₂ is industrially produced through the energy- and waste-intensive anthraquinone oxidation process, which is an indirect chemical process consisting of multiple complex steps involving the sequential hydrogenation / oxidation of anthraquinone molecules in organic solvent.² Despite its state-of-the-art large-scale production, this traditional method requires large centralized infrastructures to be economically viable, generates considerable amounts of organic waste, and involves complex separation processes to obtain high-purity H₂O₂ which presents significant sustainability challenges.⁵ Moreover, the transport, storage and handling of highly concentrated H₂O₂ is hazardous and expensive. Therefore, it is desirable to develop energy-efficient, decentralized, and low cost H₂O₂ electrosynthesis methods that would allow small-scale production at lower concentrations for onsite applications.⁶ Compared to the traditional anthraquinone oxidation process, advantages of electrochemical synthesis include utilizing renewable electricity as a low carbon emission energy source, sustainable and abundant inputs such as water and air, and mild reaction operating conditions under ambient temperature and pressure.^{1, 7, 8} Furthermore, development of electrochemical processes is beneficial for buffering intermittency issues associated with utilizing renewable energy sources by storing renewably produced electricity in the form of chemical bonds.^{2, 9}

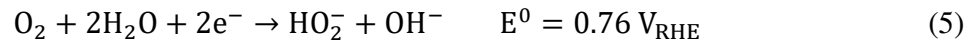
The ORR can proceed via the $4e^-$ or $2e^-$ pathway to produce H_2O or H_2O_2 , respectively. In acidic solutions, these reactions are written as follows:



where E^0 is the standard thermodynamic equilibrium potential and V_{RHE} represents the potential relative to the reversible hydrogen electrode (RHE). V_{RHE} is related to the standard hydrogen electrode (V_{SHE}) via $V_{RHE} = V_{SHE} + 0.0592 \times pH$ at 300 K. In alkaline solutions, these reactions are written as follows:



H_2O_2 is a weak acid ($pK_a = 11.63$) so at $pH > 11.63$, the H_2O_2 reaction becomes



As a result of these competing pathways and thermodynamic limitations, $2e^-$ electrocatalytic reduction of oxygen to H_2O_2 faces severe challenges in terms of limited chemical selectivity and low energy efficiency.^{10, 11} If we consider ORR proceeding via an associative mechanism, selectivity between the $2e^-$ and $4e^-$ process is determined by preference for breaking the $*-OOH$ bond (between the catalyst surface and first oxygen) or the $*O-OH$ bond (between the first and second oxygen atoms) to produce H_2O_2 or H_2O , respectively. While breaking the $*O-OH$ bond is thermodynamically more favorable than breaking the $*-OOH$ bond, a computational kinetic

analysis reveals that the rigidity of the *O-OH bond results in a higher energy barrier, thus favoring *-OOH bond cleavage.¹² Various approaches to tuning ORR selectivity involve either modifying the strength of or blocking access to the *-OOH bond; these effects can be achieved via catalyst or reaction environment tuning.

A wide range of studies have focused on enhancing efficiency and selectivity of the 2e⁻ ORR pathway by modulating parameters such as catalyst tuning, reactor design, and reaction environment conditions (electrolyte composition, pH, etc.). Among these approaches, designing highly active and selective catalysts through modification of surface and structural properties has been extensively studied.^{3, 8, 13} Metal-based catalysts, including precious metals^{10, 14-16} and metal alloys,¹⁷ have been shown to be selective for the 2e⁻ pathway with small overpotentials. However, large-scale implementation of systems relying on precious metal catalysts are constrained by the cost and supply, which has driven recent research towards investigating carbon-based materials as promising candidates for H₂O₂ production due to their low cost, abundance, and excellent stability.^{3, 18, 19} Additionally, carbon-based materials have surface and structural properties that can be tuned to induce favorable electrochemical properties¹⁵ via nanostructuring²⁰, heterodoping²¹, and incorporation of defects.¹⁹ A number of commercially available porous carbon-based materials, such as Vulcan XC-72R^{22, 23}, CMK-3^{21, 24}, carbon black^{25, 26}, and Printex L6^{22, 27}, demonstrate high H₂O₂ production rates for practical applications. Various synthesized porous carbon-based materials, including functionalized graphene²⁸, nanotubes²⁹, mesoporous carbon hollow spheres³⁰, and hierarchically porous carbon³¹, also exhibit favorability toward H₂O₂ under acidic/neutral/alkaline reaction conditions. These electrocatalysts yield a wide variety of electrochemical results, in terms of both ORR activity and

selectivity.^{3, 20, 21, 26, 32} Although great progress has been achieved improving overall H₂O₂ performance using carbon-based catalysts, the origin and mechanisms that determine selectivity and activity are still debated due to difficulty in identifying the catalytically active sites and rate determining steps.¹²

In comparison to catalyst development, considerably less work has investigated the complementary effects of reaction environment parameters. However, some insights have been identified regarding impacts of electrolyte pH on ORR mechanisms, including preference for proton coupled electron transfer (PCET) or non-coupled (non-PCET) steps (see Figure S1 for diagram of competing pathways).³³⁻³⁵ For example, previous work on a Au(100) catalyst revealed ORR onset potentials that are pH-independent on the RHE potential scale (shift ~59 mV dec⁻¹ on the SHE scale) in bulk pH>7, indicating the rate-determining step is a PCET step.³⁴ In contrast, the ORR onset potential in bulk pH<7 showed pH-dependence on the RHE potential scale suggesting a non-PCET rate-determining step. Additional studies using non-precious metal catalysts found that under both highly alkaline and acidic conditions the rate-determining-step is exclusively PCET.³⁶ Despite these studies, understanding the impact of electrolyte pH on ORR mechanisms is still challenging due to the interdependence of pH near the electrode surface (i.e. local pH) and the electrochemical reaction³⁷, which is considered less often in the field. Changes in local pH are known to influence electrocatalytic processes via modifying the charge density of an electrode which can impact adsorption energies of key reaction intermediates³⁵, altering surface speciation by influencing the site preference of reaction intermediates³⁸, or by controlling the stability of surface phases.³⁹ Building upon these studies, it is critical, yet much more difficult, to probe the interdependence of pH with other environmental parameters, especially

near the electrode surface to determine the most relevant impacts on ORR mechanisms, activity, and selectivity.⁴⁰

Some prior work has demonstrated that ORR activity and selectivity are influenced by the electrolyte environment (e.g. pH^{16, 35}, cations^{41, 42, 43}, and anions⁴⁴) near the catalyst surface, but compound effects of these parameters are still unclear, and trends observed for some model catalyst materials do not apply to other catalytic systems. At the electrode/electrolyte interface, such electrolyte species construct the electrochemical double layer (EDL), which governs the ORR activity for most model systems. For example, earlier work shows that non-covalent interactions between a hydrated cation and model Pt surface result in the stabilization of adsorbed hydroxide intermediate species for 4-electron ORR and increase with increased cation hydration energy (decreased cation radii), higher hydroxide surface coverage, and higher electrolyte pH.^{42, 43} This work postulates that hydrated cations form small clusters with adsorbed hydroxide intermediates that may block transport of reactants to the surface, with stronger interactions resulting in increased blocking and decreased ORR activity. The impacts of these non-covalent interactions are quite significant for metal catalyst surfaces such as Pt that bind strongly to ORR intermediates. In contrast, these effects are much less pronounced for carbon-based materials leading to different trends in reaction mechanism and product selectivity.^{42, 45} Further work on weaker binding catalysts suggests that influence of cations on ORR activity cannot be simply due to blocking effects of clusters formed via noncovalent interactions, but may also involve cation induced modification of reaction intermediate energies which would further impact reaction kinetics.⁴⁶ More recent investigations have been conducted for electrochemical CO₂ reduction, indicating that cations in the outer Helmholtz plane (OHP) may

induce localized electric field effects which can improve the stability of certain surface bound intermediates with strong dipole moments.^{47, 48, 49} Additionally, the different pH buffering abilities of cations has been proposed to impact product selectivity and overall reaction efficiency for CO₂ reduction⁵⁰⁻⁵², but these effects have not been systematically explored for the ORR. Collectively, these studies provide evidence that the interactions at the catalyst/electrolyte interface can impact reaction kinetics and modify the reaction mechanism, but also reveal that further insights are needed to fully elucidate these complex interactions for weaker binding material systems.

In a complementary approach to prior work in this area, we investigate the compound effects of varying potential, pH, and electrolyte cation identity on the local reaction environment and resultant ORR performance for a carbon black catalyst. We employ a rotating ring-disk electrode (RRDE) modified with an iridium oxide pH sensing probe to monitor changes in local pH during ORR under a controlled range of reaction conditions. ORR performance is evaluated for bulk pH 2-12 and four different electrolyte cations (Li⁺, Na⁺, K⁺, Cs⁺) which cover a range of different cation sizes. For near-neutral bulk pH electrolytes, we observe large changes in local pH with increasing ORR current density; we also find that different electrolyte cations provide variable local pH buffering at these conditions, modulating the onset and overall magnitude of local pH changes. By coupling these results with key ORR performance metrics, such as onset potential, selectivity, and Tafel slopes, we find that changes in local pH more strongly dictate changes in ORR mechanisms and performance, while the cation identity appears to modulate local pH to influence these trends. Uncovering this relationship between local and bulk reaction

environments will further improve our understanding of how bulk operating parameters impact electrocatalytic processes.

2. Experimental Section

2.1 Materials and Chemicals

Mesoporous carbon CMK-3 was purchased from ACS Material and used as received. The electrolytes used in the H₂O experiments were prepared from mixtures of 0.1 M H₂SO₄, 0.1 M MOH, and 0.1 M M₂SO₄ where M⁺ = Li⁺, Na⁺, K⁺, Cs⁺ (99.999% trace metals basis, ThermoFisher Scientific) using ultrapure de-ionized H₂O (Milli-Q system, Millipore R=18 MΩ cm). The iridium oxide (IrO_x) nanoparticle solution for the Au-IrO_x ring was prepared from iridium tetrachloride (Strem Chemicals Inc.), 30 wt% hydrogen peroxide (Sigma Aldrich), anhydrous sodium carbonate (Sigma Aldrich), and anhydrous oxalic acid (Sigma Aldrich). All gases used (O₂ and N₂) in this work had a purity of >99.995% (Praxair, Inc.). Standard pH electrolyte solutions (pH 2,4,6,9,10,12) used to calibrate the iridium oxide (IrO_x) Au ring electrode were prepared from mixtures of 0.1 M H₂SO₄, 0.1 M KOH, and 0.1 M K₂SO₄; pH was confirmed using a pH meter (Fisherbrand Accumet AB150).

2.2 Electrode Preparation

The disk working electrode was prepared by mixing 2 mg of CMK-3, 197.3 μl of ethyl alcohol (ThermoFisher Scientific) and 2.7 μl of cation-exchanged Nafion solution (Sigma-Aldrich). The resultant catalyst ink was sonicated for three hours. 10 μl of the catalyst ink was then drop-casted onto a clean and polished glassy carbon (GC, 0.196 cm²) disk (Sigradur G Hochtemperatur-Werkstoffe GmbH) to obtain a catalyst loading of 0.5 mg cm⁻². Prior to

electrodeposition, the Au ring RRDE assembly (PINE Research Instrumentation) was polished using a 0.05 μm alumina slurry (Pine Research Instrumentation), rinsed with Millipore H_2O , and sonicated for five minutes.

2.3 Electrodeposition of Iridium Oxide on the Au Ring

A ligand-free suspension of colloidal IrO_x nanoparticles was synthesized using a documented approach.⁵³ IrO_x was electrochemically deposited on the Au ring of the RRDE (Figure S2) using a three-electrode configuration equipped with a potentiostat (Biologic VSP 200), a platinum (Pt) wire counter electrode, and a Ag/AgCl (sat'd KCl) reference electrode. The electrodeposition was performed via cycling potential between -0.6V to 0.5V vs. Ag/AgCl for 300 cycles at a scan rate of 1 V s^{-1} .⁵⁴ Following the deposition, the entire RRDE shaft was stored in ultrapure de-ionized H_2O overnight to stabilize the catalyst surface, resulting in consistent open circuit potential (OCP) measurements and improved response time to pH changes. The IrO_x -Au ring pH response was tested using a series of prepared standard pH electrolytes (pH 2-12) and measuring the corresponding changes in OCP.

2.4 Electrochemical Measurements

The oxygen reduction performance was evaluated in a four-electrode configuration using an RRDE controlled with a bipotentiostat (Biologic VSP 200). An electrochemical heart shaped cell (Pine Research Instrumentation) was equipped with a gas dispersion tube, a saturated silver/silver chloride (Ag/AgCl sat'd KCl) reference electrode, a Pt wire counter electrode, and a GC disk with deposited catalyst and a Pt ring as working electrodes. Prior to the ORR measurements, the Pt ring electrode was cleaned by successive cycle voltammetry (CV) in N_2 -

saturated electrolyte within the potential range from 0.05V and 1.2V vs. RHE at 20 mV s⁻¹ until steady CVs were obtained. Then, background current (capacitive current) was collected between 0.8V to 1.0V vs. RHE. Electrochemical impedance spectroscopy (EIS) was then performed to measure the uncompensated resistance (R) of the liquid electrolyte for the subsequent IR correction. The potential scale for different composition electrolytes was calibrated to the reversible hydrogen electrode (RHE) using a Pt wire (Sigma-Aldrich) as the working and counter electrode in H₂-saturated electrolyte.

After purging with O₂ for 30 minutes, linear sweep voltammetry (LSV) measurements were carried out between 1.1V and 0.05V vs. RHE at 20 mV s⁻¹ with a rotation speed of 1600 rpm. To obtain faradaic currents on the disk electrode, the capacitive current is subtracted from the measured current. The amount of H₂O₂ produced on the disk was quantified by holding the Pt ring at 1.2 V vs. RHE to oxidize the hydrogen peroxide species formed on the disc electrode. The system's ring collection efficiency (*N*) was determined to be 0.224 using the reversible [Fe(CN)₆]^{4-/3-} redox couple (+0.36 vs. SHE). The selectivity of H₂O₂ and the number of electrons transferred (*n*) were calculated according to the following equations [Eqns. (1)-(2)]:⁵⁵

$$\text{H}_2\text{O}_2(\%) = 200 * \frac{\frac{I_R}{N}}{|I_D| + \frac{I_R}{N}} \quad (1)$$

$$n = \frac{4 *}{|I_D|}$$

where *I_R* is the ring current and *I_D* is the disk current.

To detect local pH changes during ORR, chronopotentiometry (CP) was conducted on a CMK-3 disk catalyst using the IrO_x-Au ring RRDE assembly described previously. Before and after each measurement, the OCP-pH response of the IrO_x-Au ring was tested in the respective

bulk pH electrolyte to ensure stability and reliability of the collected data. If the ring OCP measurements were unstable, the ring was conditioned by applying -0.256 V vs. RHE for 2 minutes to stabilize the IrO_x oxidation state; this conditioning has been previously shown to positively influence pH sensitivity of these probes.⁵⁶ The CP protocol consisted of applying five minute constant current holds to the CMK-3 disk electrode (-0.02,-0.2,-0.4,-0.6,-0.65 mA) while simultaneously measuring the IrO_x-Au ring OCP. Using a calibration curve, the local pH was calculated from the ring OCP averaged over the last three minutes. All measurements were performed in O₂-saturated 0.1 M (M/H)₂SO₄ electrolytes at varying pH (2-12) at a scan rate 20 mV s⁻¹ with 1600 rpm and the CMK-3 catalyst loading amount on the disk electrode was set to 0.5 mg cm⁻².

The potential with respect to the Ag/AgCl reference electrode can be converted to that of the reversible hydrogen electrode (RHE) by the following Equation 3:

$$E(\text{V vs. RHE}) = E\left(\text{V vs. } \frac{\text{Ag}}{\text{AgCl}}\right) + 0.197(\text{V}) + 0.059 * \text{pH} \quad (3)$$

Tafel slopes ($\frac{\alpha F}{2.3RT} \equiv b$) were calculated by the Tafel equation (Eqn. 4) obtained from the linear relation in the overpotential (η) vs. log kinetic disk current density ($j_{\text{disk},\text{kin}}$).

$$\eta = \frac{2.3RT}{\alpha F} * \log(j_0) + b * \log(j_{\text{disk},\text{kin}}) \quad (4)$$

Here, α , F , R , T , and j_0 refer to the transfer coefficient, Faraday's constant (96489 C mol⁻¹), gas constant (8.314 J mol⁻¹), temperature (298 K), exchange current densities, respectively. The

kinetic disk current densities ($j_{disk,kin}$) can be determined according to the Koutecky-Levich equation (Eqn. 5):⁵⁷

$$\frac{1}{j_{disk}} = \frac{1}{j_{disk,kin}} + \frac{1}{j_{disk,lim}} \quad (5)$$

where j_{disk} and $j_{disk,lim}$ are the measured current density and the diffusion-limited current density, respectively.⁵⁸ In addition, $j_{disk,lim}$ can be calculated using the Levich equation (Eqn. 6):

$$j_{disk,lim} = 0.62nFD^{\frac{2}{3}}v^{-\frac{1}{6}}\omega^{\frac{1}{2}}C \quad (6)$$

where n , F , C , D , v and ω refer to the number of electrons transferred, the Faraday's constant (96485 C mol⁻¹), bulk concentration of oxygen (1.2×10^{-6} mol cm⁻³ for pH 5-13 and 1.1×10^{-6} mol cm⁻³ for pH 2), the diffusion coefficient of oxygen (1.9×10^{-5} cm² s⁻¹ for pH 5-13 and 1.8×10^{-5} cm² s⁻¹ for pH 2), the kinetic viscosity (0.01 cm² s⁻¹), and the angular velocity of the disk ($\omega = 2\pi N$, N is the linear rotation speed).^{21, 59}

3. Results and Discussion

3.1. Measuring Local pH for ORR

To detect local pH changes under well-defined mass transport conditions, we use an RRDE technique employing a Au ring modified with a potentiometric pH sensing probe material, iridium oxide (IrO_x).^{54, 60} The open circuit potential (OCP) of the IrO_x closely follows super-Nernstian behavior, shifting by approximately 60-90 mV dec⁻¹ in response to the pH of its immediate environment.⁶¹ This experimental platform enables us to monitor OCP of the IrO_x-Au ring, correlate this value to a calibration curve, and approximate the pH near the disk (i.e.

catalyst surface). The IrO_x-Au ring pH response was calibrated using a series of prepared standard pH electrolytes (pH 2-12). This technique provided consistent results with a pH response of 77 mV dec⁻¹ and exhibited a good linear fit ($R^2=0.995$) between OCP and known pH when the IrO_x-Au ring was allowed to stabilize for five minutes before making a final reading (Figure 1A). These results demonstrate the ability of the IrO_x-Au ring electrode to detect changes in local pH as the rotation brings the local environment of the disk to the ring.

To investigate how different reaction rates affect local pH during ORR, chronopotentiometry (CP) measurements were performed on a carbon black (CMK-3) catalyst disk electrode for five minutes while simultaneously measuring the OCP of the IrO_x-Au ring electrode. Before and after each measurement, the OCP-pH response of the IrO_x-Au ring was tested in the respective bulk pH electrolyte to ensure stability and reliability of the collected data. Using the measured OCP of the IrO_x-Au ring and the calibration curve in Figure 1A, the local pH during ORR was calculated as a function of disk current density (Figure 1B). These measurements were performed in a series of electrolytes with varying bulk pH, while maintaining consistent salinity or total ion concentration (i.e. 0.1 M (K/H)₂SO₄, with an anion concentration of 0.1 M [SO₄²⁻] and a total cation concentration of 0.2 M [H⁺] and [K⁺]). Sulfate was chosen as the anion in this study since the application for our work is production of H₂O₂, which can be used as an oxidant for environmental remediation, and therefore, it is important to operate without perchlorate ions since they are well known as persistent environmental pollutants.⁶² In general, as the magnitude of the disk current density increases, local pH near the catalyst surface becomes more alkaline, as is expected because higher ORR currents lead to faster proton consumption or hydroxide formation. Bulk pH electrolytes with the greatest

concentrations of protons (pH 2) or hydroxide ions (pH 10 & 12), exhibit only a small change in local pH. However, a more complicated behavior is observed for the neutral bulk pH electrolytes. Bulk pH 4 demonstrates the greatest change in local pH across the range of currents examined ($0 - 3.25 \text{ mA cm}^{-2}$). Bulk pH 6 similarly exhibits a large change in local pH, but the onset of this change occurs at much lower currents compared to a bulk pH of 4. These results reveal that bulk pH 4 and 6 support local conditions closer to neutral/alkaline under higher ORR current densities, which is a critical property to consider when evaluating ORR performance.

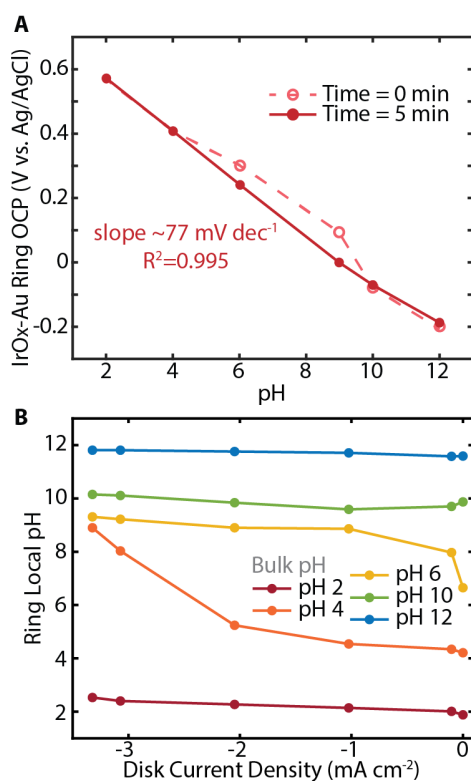


Figure 1. A) Calibration curve for IrO_x-Au ring OCP in standard pH electrolytes. B) Change in local pH detected by ring as a function of ORR current density on the disk from CP measurements conducted via RRDE with IrO_x-Au ring and CMK-3 catalyst disk. All

measurements performed in O₂-saturated 0.1 M (K/H)₂SO₄ electrolyte with varying bulk pH between 2-12 at a rotation rate of 1600 rpm and catalyst loading of 0.5 mg cm⁻² on the disk electrode.

3.2. pH Effects on ORR Performance

To draw connections between potential, pH, and electrocatalytic performance metrics, we investigated ORR on mesoporous carbon black (CMK-3) using an RRDE equipped with a Pt ring held at 1.2 V vs. RHE for H₂O₂ detection.¹⁵ As a representative data set, Figure 2A shows linear sweep voltammetry (LSV) curves for a set of 0.1 M (K/H)₂SO₄ electrolytes spanning a range of bulk pH values. Similar to prior work, we experimentally define onset potential as the potential to reach a ring current density of 0.01 mA cm⁻².²¹ We find there is an anodic shift in the H₂O₂ onset potential with increasing pH (Figure 2A). Previous literature has indicated that reactions involving a sequence of proton and electron transfer steps in series (non-PCET), rather than in a concerted fashion (PCET), will have a thermodynamic overpotential with a strong dependence on pH, as can be observed in our data.⁶³ Other observations which may explain this trend in onset potential are based on the pH-dependent adsorption strength of O₂ and key reaction intermediates¹⁶, as well as, a recent theoretical study that reports the applied electric field becomes more negative with increasing pH, which influences the adsorption of key ORR intermediates on weak binding surfaces, such as carbon.³⁵ Figure 2A also reveals that, after reaching a maximum value, the ring current density decreases gradually with increasing electrode potential, reflecting differences in the production and/or decomposition mechanisms of H₂O₂ in different pH environments.⁶⁴

Additionally, we investigate potential changes in the ORR mechanism by evaluating Tafel slopes (b) using the kinetic disk current density ($j_{\text{disk,kin}}$) from the LSV results and the Tafel equation (Eqn. 7):

$$\eta = a + b * \log(j_{\text{disk,kin}}) \quad (7)$$

where η is the overpotential, b is the Tafel slope, a is $(2.303RT/F) * j_0$, j_0 is the exchange current density, R is the gas constant ($8.314 \text{ J K}^{-1} \text{ mol}^{-1}$), T is temperature (298 K), and F is Faraday's constant (96485 C mol^{-1}). Tafel slopes are labeled in Figure 2B, revealing a significant jump from 132 mV dec^{-1} at bulk pH 2, compared to Tafel slopes of $46\text{-}67 \text{ mV dec}^{-1}$ for bulk pH 4-12. Noting the significant difference in Tafel slope between a bulk pH of 2 and 4, we find that local pH may provide more relevant insight to explain these and other complex ORR activity and selectivity trends in varying reaction conditions. To enable comparison with reported systems that primarily analyze Tafel slopes over a range of currents for a bulk pH value, we similarly report Tafel slopes calculated using kinetic disk current densities in the range of $1\text{-}3 \text{ mA cm}^{-2}$ at each bulk pH, while considering average local pH conditions. Bulk pH 4 and 6 support local conditions that are much closer to neutral/alkaline at kinetic current densities (average local pH 5.2 and 9, respectively) and therefore present similar Tafel slopes as the higher bulk pH conditions. A higher Tafel slope in acidic conditions indicates a less efficient electrochemical process, which is consistent with the observed later onset potential and lower total disk current density (Figure 2A). The Tafel slopes for the neutral/alkaline local conditions are all very similar, implying similar ORR kinetics. These changes in Tafel slopes indicate a pronounced change in the ORR mechanism and/or rate determining step as a function of electrolyte pH, despite using an identical catalyst for all measurements.⁶⁵

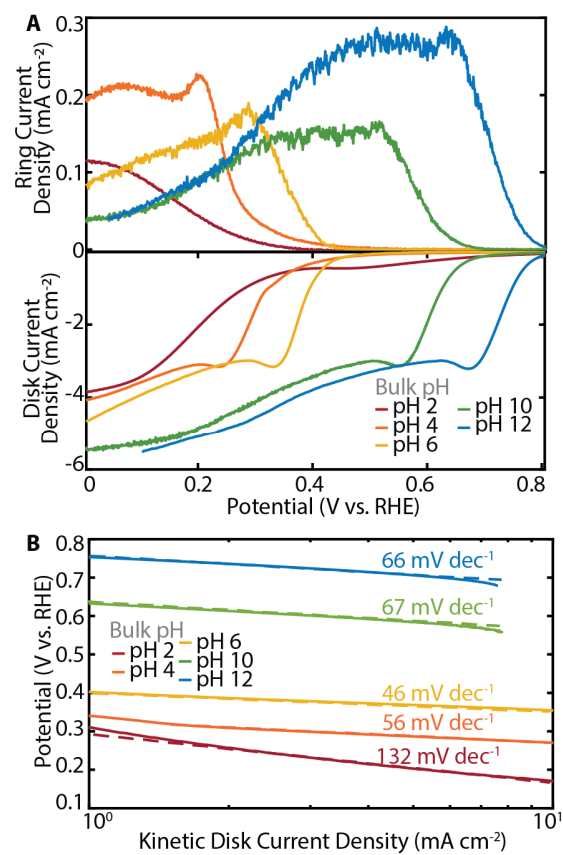


Figure 2. A) LSVs of RRDE measurements with CMK-3 catalyst disk and Pt ring to evaluate ORR performance and product selectivity as a function of electrolyte bulk pH and electrode potential. 2nd LSV is shown for all measurements. Current is collected on the Pt ring at a constant potential of 1.2 V vs. RHE. B) Tafel plot with labelled Tafel slopes. Note that bulk pH conditions of 2, 4, 6, 10, and 12 support average local pH conditions at these kinetic current densities of 2.2, 5.2, 9, 10, and 12, respectively. All measurements performed in O₂-saturated 0.1 M (K/H)₂SO₄ electrolytes with varying bulk pH with a scan rate 20 mV s⁻¹, rotation rate of 1600 rpm, and catalyst loading of 0.5 mg cm⁻².

As we further consider the analysis of ORR performance metrics vs. local pH trends, we revisit the H₂O₂ onset potentials, which are shown on both an RHE and SHE scale vs. local pH in Figure 3A. At the onset current density, the rate of proton consumption / hydroxide formation is very low, such that local pH does not deviate much or at all from the bulk pH, except for bulk pH 6 which already measures closer to a local pH of ~7.9. Observing the H₂O₂ onset potential trends for local pH >7, the onset of ORR does not shift significantly with pH on an SHE scale, but shifts by ~95 mV per local pH on an RHE scale (Table S1-S3). Based on this observation, we postulate neutral/alkaline local pH conditions support non-PCET mechanisms, as is common for many catalysts that bind oxygen weakly and favor 2-electron ORR.^{63, 66} However, below pH 7, there is a greater shift in onset potential on an SHE scale and closer to 0 shift on an RHE scale, which suggests PCET rate determining steps or overall a complicated mechanism scheme. We can also compare the observations for H₂O₂ onset potentials and Tafel slopes, where we observed a Tafel slope of 132 mV dec⁻¹ in acidic local pH conditions (bulk pH 2), which was previously reported to be associated with PCET step mechanisms.⁶⁷ In contrast, we found Tafel

slopes of 46-67 mV dec⁻¹ in neutral/alkaline local pH conditions (bulk pH 4-12), previously attributed to a non-PCET step mechanisms. We highlight that the bulk pH 4 condition presents an interesting case, due to the large change in local pH measured at the onset vs. kinetic current regimes. At onset current density, the catalyst experiences a local pH of 4, yet experiences a much higher local pH of 6-8 in the kinetic current regime used to calculate a Tafel slope (Figure 1B). This change in environment is paralleled by our observations of a likely PCET rate determining step, similar to bulk pH 2 based on the onset current (Figure 3A), but a Tafel slope that more closely matches those of the alkaline conditions, based on kinetic current (Figure 2B). This may indicate a change in reaction mechanism over the range of currents examined in a 0.1 M (K/H)₂SO₄ environment with bulk pH 4.

To further explore the impact of pH and potential on electrocatalytic performance and possible reaction mechanisms, we analyze trends in the maximum H₂O₂ current (ring current) densities and selectivities. Each condition was measured 3 times, and the average values and standard deviations for this metric at each condition are included in Figure S8. For the 0.1 M (K/H)₂SO₄ system, the maximum ring current density was always greatest for bulk pH 12, with more moderate maximum ring currents for bulk pH 4, 6, and 10, and the lowest maximum ring current was always observed for bulk pH 2. Using both the ring and disk current densities, we also calculated the H₂O₂ selectivities as a function of the disk potential (Figure 3B). Closely following the trends in maximum ring current density, the highest H₂O₂ selectivity is consistently observed for bulk pH 12, with more moderate selectivities for bulk pH 4, 6, and 10, and the lowest selectivity for bulk pH 2. Figure 3C highlights the maximum H₂O₂ selectivity for each condition plotted as a function of both bulk and local pH, determined at the respective operating

currents. Overall, we find that the maximum H_2O_2 selectivity increases with increasing pH, but within the bulk pH range of 4-10, the selectivity is relatively constant. Replotting against local pH, we see that the selectivity trend at intermediate bulk pH does not derive from an insensitivity to pH, but rather than all three conditions have relatively similar local pH. We posit that explaining selectivity trends with respect to local pH is more relevant than considering bulk pH, as the local conditions are a more accurate representation of the actual catalyst environment.

The selectivity trends for electrolytes with varying pH appear to agree well with previous work that demonstrates ORR activity dependence on defects in carbon-based catalysts.^{26, 68} These defect sites are surrounded by carbon atoms passivated by adsorbed H. In acidic conditions, protons in the electrolyte will be sterically blocked from accessing the first O (directly adsorbed on the catalyst surface) due to the neighboring surface-adsorbed H atoms. Therefore, protons have stronger affinity for the second O atom, resulting in lower H_2O_2 selectivity on carbon catalysts in acidic conditions.¹² This also correlates well with observations of nitrogen-doped carbons⁶⁹ and transition metal-N-C catalysts⁷⁰ where the various dopants introduce additional defects sites leading to degraded catalytic activity upon protonation, possibly via subsequent anion binding on neighboring protonated sites. Our results suggest that there may be a similar effect in undoped carbon-based catalysts via protonation of carbon atoms surrounding defect active sites.

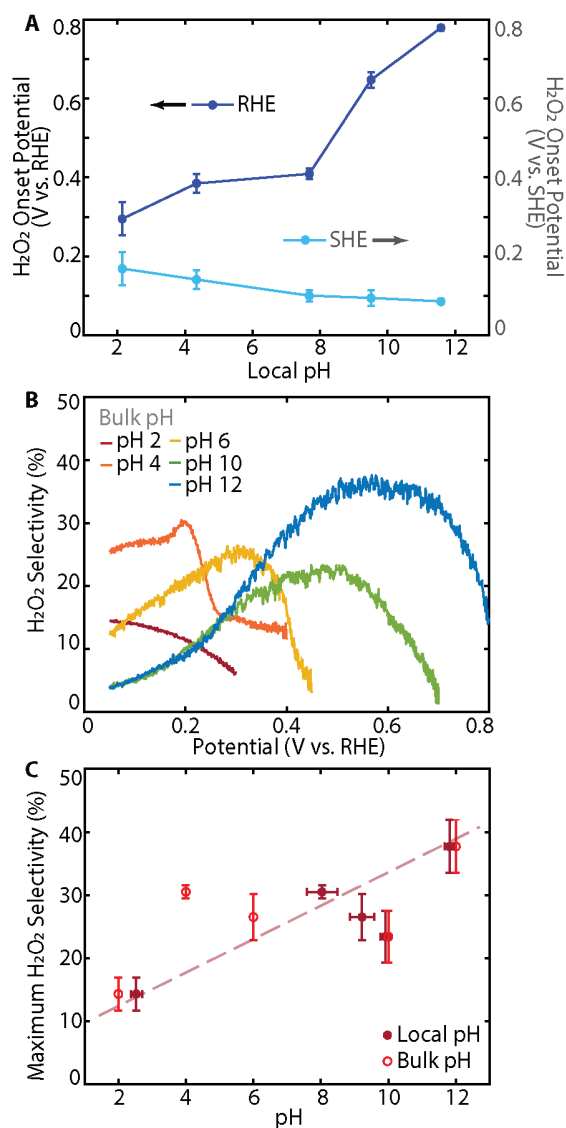


Figure 3. A) H_2O_2 onset potential as a function of local pH displayed vs. RHE (left axis) and vs. SHE (right axis). Onset potential is defined as the potential when ring current reaches 0.01 mA cm^{-2} . B) H_2O_2 selectivity (%) as a function of electrode potential. C) Calculated maximum H_2O_2 selectivity (%) as a function of bulk pH and local pH during ORR. Dotted line included to guide the eye. Data points and error bars reflect average and one standard deviation, respectively, for

three measurements at each condition. When not visible, error bars are very small and overlap with data point.

3.3. Cation Effects on ORR Performance

Prior work has established that interactions between electrolyte cations and the electrocatalytic interface significantly influence the activity and selectivity of electrochemical reactions, but there are still longstanding questions regarding the mechanisms of these observed effects and the interplay between cation identity and pH variation, particularly for the ORR.^{42, 45, 46, 48, 50-52, 71, 72} To address these questions, we first examined changes in local pH during ORR in electrolytes spanning the same range of bulk pH and maintaining consistent salinity, but with varied alkali metal cations (i.e. 0.1 M (M/H)₂SO₄ where M⁺ = Li⁺, Na⁺, K⁺, Cs⁺) over a range of ORR current densities (0 – 3.25 mA cm⁻²) at the disk (Figure S3). The magnitude of the local pH change across the lowest to highest ORR currents measured for the different alkali metal cations are shown in Figure 4A. In acidic electrolyte (bulk pH 2), the changes in local pH were all within 0.5 – 1 pH unit. The smallest radii cation (with the largest hydration shell radii, Li⁺) was least effective at buffering and exhibited the greatest change in local pH, while the larger cations (smaller hydration shells) followed a trend in local pH change of Li⁺>Na⁺>K⁺>Cs⁺, such that larger cations provided slightly enhanced buffering of the local pH. In alkaline electrolytes (bulk pH 10 and 12), the changes in local pH are again mostly within 0.5 – 1 pH unit, but we observed no clear trend with respect to cation identity. In contrast, the greatest overall change in local pH was observed in neutral or slightly acidic bulk pH electrolytes (bulk pH 4 and 6). Interestingly, at a bulk pH of 4, Cs⁺ is still able to provide reasonable buffering for the local pH (change of only

~1.5 pH units), while electrolytes with the other cations experience significantly larger local pH changes (change of 4-5 pH units). At a bulk pH of 6, all cations exhibit similar buffering capabilities, with local pH change on the order of ~2.5-3.5 pH units. Overall, we note that Cs⁺ is least effective at buffering local pH (across the full range of currents measured) in electrolyte with bulk pH 6, while all smaller cations are least effective in buffering local pH in electrolyte with bulk pH 4. For increasingly basic or acidic bulk pH conditions (higher or lower pH values), all cations exhibit increasingly improved buffering capabilities.

We can examine these trends more closely by considering the magnitude of current that triggers these large changes in local pH at neutral bulk pH conditions as shown in Figure S3. At bulk pH 4, electrolytes with Li⁺ and Na⁺ experience change in local pH at very low ORR current densities (~0.1 mA cm⁻²), while electrolytes with K⁺ and Cs⁺ do not experience a significant change in local pH until after higher ORR current densities have been reached (2.0 – 3.0 mA cm⁻²). At pH 6, electrolytes with Li⁺, Na⁺, and K⁺ similarly experience change in local pH at very low ORR current densities. Electrolyte with Cs⁺ at bulk pH 6 does not reflect a large change in local pH until ~1 mA cm⁻² of ORR current is reached.

We hypothesize and explain these trends considering a range of electrostatic effects between electrolyte cations of varying radii localized within the electrochemical double layer and water molecules or the cathodic catalyst surface (Figure 4B).^{46, 48, 50-52} We first note that prior studies of ORR provide evidence that electrolyte cations remain at least partially solvated at the cathode/electrolyte interface and Frumkin's theory of cation surface adsorption is not likely to be applicable under operating potentials for ORR.^{45, 72, 73} Moreover, many of these previous reports

studied model catalyst systems, such as Pt, which is known to behave quite differently from our weaker binding carbon-based catalyst.⁷⁴ Next, we consider that the pKa of hydrolysis for hydrated cations in the bulk electrolyte is inversely proportional to the electrostatic energy of interaction between the O atoms of water molecules and the respective cation, such that the pKa increases with increasing cation radii.^{51, 75} However, this trend is reversed and the pKa of hydrated cations near polarized cathodes decreases with increasing cation radii due to increased electrostatic interaction of larger cations with the cathode surface. If the pKa of a hydrated cation is lower than the local pH at the cathode surface, one of the hydrating water molecules will dissociate and release a proton. This enables hydrated cations to buffer electrolyte near cathode surfaces, and indicates that cations with larger radii, such as Cs⁺, will act as more effective buffers at lower pH. Similar to these prior studies, we postulate that this phenomenon also explains observations in our system, where Cs⁺ is significantly more effective at buffering the local pH for bulk pH 4, compared to all smaller cations in this study. This buffering effect is further magnified due to the energetic favorability for cations with larger radii to localize in higher concentrations at the outer Helmholtz plane (OHP) of the electrochemical double layer (typically <1 nm from the cathode surface).⁴⁸ We hypothesize a higher concentration of Cs⁺ cations exists at the OHP compared to smaller cations, in addition to its enhanced buffering capabilities at lower pH, which would lead to smaller changes in local pH for Cs⁺ electrolyte at bulk pH 4 and lower, as is observed here.

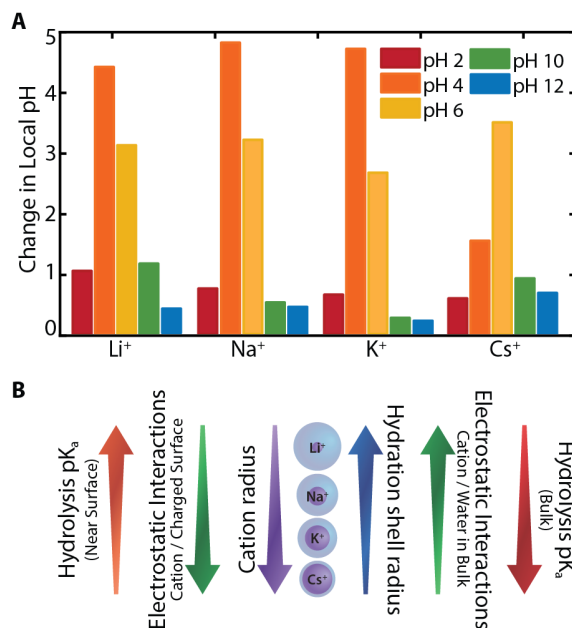


Figure 4. A) Change in local pH from zero (0 mA cm⁻²) to high (3.25 mA cm⁻²) ORR currents as a function of bulk pH for different 0.1 M (M/H)₂SO₄ electrolytes where M⁺ = Li⁺, Na⁺, K⁺, Cs⁺. Change in local pH calculated from CP experiments. B) Illustration summarizing relative cation properties that may drive observed trends.

We further examined the impact of cation identity on activity and selectivity toward H₂O₂ (Figure S4 and S5), revisiting Tafel slopes, onset potentials, and maximum H₂O₂ selectivity as performance metrics. Tafel slopes were again calculated for each bulk pH condition and interpreted noting the average local pH at the kinetic current densities for each system to investigate the influence of cation identity on the ORR kinetics (Figures 5A and S6). We again find that there is a significant difference in Tafel slopes between acidic and neutral/alkaline local pH conditions, but there is no substantial dependence on cation identity. We do note that electrolytes with Cs⁺ at higher pH generally have a lower Tafel slope, but these results do not indicate an extended trend across all cations examined. Tafel slopes for acidic local conditions

were found to be between 108 mV dec⁻¹ to 140 mV dec⁻¹ while Tafel slopes for neutral/alkaline local conditions were between 40 mV dec⁻¹ to 90 mV dec⁻¹. Overall, based on the differences in Tafel slopes, we hypothesize that the cation identity indirectly impacts the ORR mechanism by modulating the overall impact of pH.

In Figures 5B and S7, H₂O₂ onset potential is plotted as a function of local pH for the different cations. Figure 5B reveals that all electrolytes exhibit similar behavior at high pH, but the H₂O₂ onset potential vs. SHE for Cs⁺ and K⁺ electrolytes bifurcates from Na⁺ and Li⁺ electrolytes at lower pH. Therefore, at these low onset currents, we predict rate determining steps involving non-PCET steps for all electrolytes at higher pH values, as well as Na⁺ and Li⁺ electrolytes at low pH. Conversely, we hypothesize that K⁺ and Cs⁺ electrolytes at lower pH values support PCET steps or more complex mechanisms schemes where the onset potential does shift on an SHE scale, but not at an expected value of ~59 mV per pH.

H₂O₂ selectivity similarly highlights a bifurcation of performance for Cs⁺ or K⁺ and Na⁺ or Li⁺ electrolytes (Figures 5C and S9). At higher local pH values (> pH 8), we note that H₂O₂ selectivity for Na⁺ and Li⁺ electrolytes is mostly quite poor at 5-15%, while H₂O₂ selectivity for Cs⁺ and K⁺ electrolytes are much higher at 20-40%. At low pH, the H₂O₂ selectivity for all electrolytes is quite poor (<15%); we do observe somewhat higher selectivity for Cs⁺ and K⁺ vs. Na⁺ and Li⁺, but the difference is less pronounced than at higher pH.

The overall lower ORR activity and decreased H₂O₂ selectivity for Li⁺ and Na⁺ electrolytes may be impacted by variability in pH buffering effects, as described previously, but may also involve various other effects as have been observed for ORR or other cathodic reactions in

previous literature. As noted above, larger cations are predicted to localize in increased concentrations at the OHP; their presence may further influence product selectivity due to electrostatic interactions between these solvated cations and adsorbed reaction intermediates that have large dipole moments or are easily polarizable.^{48, 50} Tuning relative stability of adsorbed OOH species would impact favorability for 2- vs. 4-electron ORR processes. In addition, steric effects due to varied cation radii could affect the OHP potential for ORR, as has been investigated previously using a modified Poisson–Boltzmann equation.⁷⁶ We postulate these effects may also explain the observed differences in the ORR mechanism and performance metrics for Li⁺ and Na⁺ electrolytes. Similarly, to explain the 2e⁻ selectivity enhancement in K⁺ and Cs⁺, we hypothesize there exists favorable interactions of the larger cations at or near the electrode surface with the intermediate adsorbates and/or active sites, as has been observed in recent CO₂ reduction literature. Such interactions might include cations in the outer Helmholtz plane generating a local electric field that can improve the stability of the key ORR intermediates (e.g. *OH, *OOH, and *O)^{42, 49, 50} and/or other non-covalent interactions between the cation and the oxygen-adsorbates.⁷⁷

In short, we provide evidence that cations play an important role indirectly modifying overall ORR performance via changes in local pH through surface or near-surface interactions in the electrochemical double layer. Our results further support previously proposed pH-buffering effects related to the pK_a of hydrolysis for hydrated cations near cathode surfaces which influence local pH, pH-dependent reaction mechanisms, and proton transport.^{51, 52} These local environment effects are critical to describe and understand observed trends in ORR activity and selectivity; continued work in this area will be critical to uncover compound influences of these

and other local environment properties to drive continued improvement of sustainable electrocatalytic technologies.

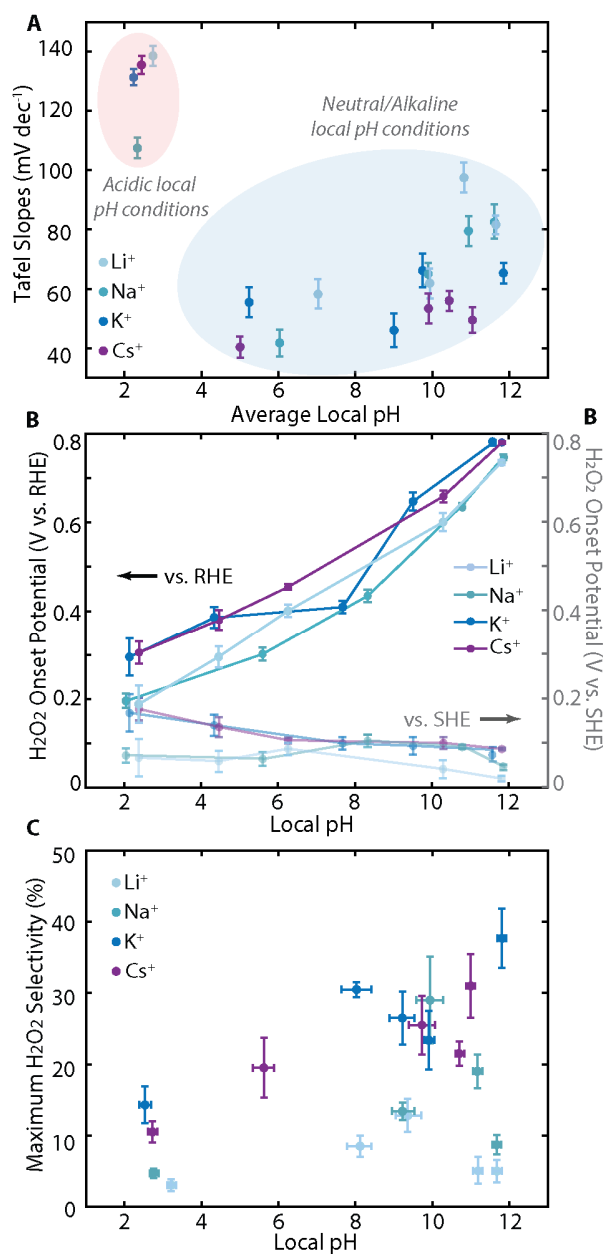


Figure 5. A) Tafel slopes (mV dec^{-1}) as a function of local pH evaluated between 1-3 mA cm^{-2} disk kinetic current density, B) H_2O_2 onset potential displayed vs. RHE (left axis) and vs. SHE

(right axis), and C) calculated maximum H₂O₂ selectivity (%) for different cations. Data points and error bars reflect average and one standard deviation, respectively, for three measurements at each condition.

4. Conclusions

In this work, we demonstrate that tuning bulk reaction environment parameters, such as electrolyte ion composition and electrode potential, provides insights into how we can control the catalyst's local reaction environment, and in turn, dictate its activity and selectivity towards H₂O₂. We also demonstrate the ability to experimentally measure changes in local pH under a carefully controlled range of ORR conditions and relate these measurements to the observed differences in ORR mechanisms and performance metrics for a carbon black CMK-3 catalyst. By combining local pH measurements, ORR electrochemical testing, and Tafel analysis, we gain insight into the possible pH and cation effects at or near the catalyst surface that impact ORR performance. From these findings and building off of previous work, we hypothesize possible interactions in the local catalyst environment that explain the observed trends in ORR performance.

Relative changes in local pH were greatest in neutral bulk pH electrolytes (bulk pH 4 and 6) across a range of operating potentials and ORR currents. In addition, different alkali metal cations provide variable pH buffering, with Cs⁺ providing the greatest buffering at bulk pH 4, which thereby modulates the onset and overall magnitude of local pH change as a function of ORR current. By coupling these results and local environment probes with key ORR performance metrics, such as onset potential, selectivity, and Tafel slopes, we can better explain

the observed trends in varying reaction conditions. Tafel slopes reveal a significant influence of pH on the ORR mechanism with higher Tafel slopes ($>108 \text{ mV dec}^{-1}$) observed in more acidic local conditions and lower Tafel slopes observed for more neutral/alkaline local conditions ($\sim 66 \text{ mV dec}^{-1}$). Through mechanistic and kinetic studies, we find that acidic local environments support PCET rate determining steps, which agrees with the observed higher Tafel slopes. In contrast, more neutral/alkaline local conditions support non-PCET rate determining steps. H_2O_2 selectivity generally increases with local pH for larger cations (K^+ , Cs^+) while smaller cations (Li^+ , Na^+) demonstrate lower H_2O_2 selectivity at higher local pH conditions. We hypothesize and explain the dependence of ORR performance on cation identity via their variable pH-buffering abilities due to the differences in the cation radii, hydration shells, and electrostatic interactions near a polarized cathode surface. We find that changes in local pH caused by bulk pH and reaction rate or current have a stronger and more direct impact on ORR mechanisms and performance for CMK-3, while the cation identity indirectly impacts performance trends by more weakly modulating local pH changes. Overall, this work demonstrates and utilizes a platform for systematically analyzing the separate and compound impacts of bulk reaction environment parameters on local reaction environments and improves our fundamental understanding of how these parameters impact reaction mechanisms and electrocatalytic performance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online.

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Bulk Layer
Boundary Layer
Double Layer
Catalyst

